



Analysis of Roughness Effects in Laminar Fluid Flow by Numerical Methods

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Contents:

- 1.1 [Abstract](#)
- 2.1 [Introduction](#)
- 3.1 [Prior Work & Results](#)
- 4.1 [Problem Formulation](#)
- 5.1 [Numerical Analysis](#)
- 6.1 [Results & Discussions](#)
- 6.5 [Extended Results](#)
- 7.1 [Path Forward](#)
- 8.1 [Conclusion](#)
- 9.1 [Self-Assessment](#)
- 10.1 [References](#)

1.1 Abstract

Numerical investigations are conducted to examine the influence of wall roughness on fully developed laminar pipe flow. The roughness consists of square bars with varying size and pitch. The findings reveal that, despite the presence of rib height and pitch variations, the established inverse relationship between the friction factor and Reynolds number in smooth pipes persists in rough pipes. The friction factor is observed to vary quadratically with roughness height and linearly with roughness pitch at a given Reynolds number. A unified correlation for the friction factor is proposed, successfully collapsing the data.

Keywords: Laminar, Effective Reynolds Number, Friction Factor, Roughness

2.1 Introduction

This research aims to extend our understanding of fully developed laminar pipe flow by investigating the influence of wall roughness through comprehensive numerical simulations. Our exploration builds upon the classical work of *Nikuradse*, who laid the foundation for the analysis of friction factors in pipes, particularly in the context of artificially roughened surfaces. Nikuradse's experiments identified three distinct flow regimes: laminar, transitional, and fully turbulent. Notably, in fully turbulent flows, roughness played a significant role once it exceeded a critical size. However, the impact of roughness in laminar flows has been a focal point within the microscale flow community, revealing substantial deviations from Nikuradse's initial proposals.

In our study, we specifically investigate the effects of rough surfaces, characterized by square bars of varying size and pitch, on laminar flows. By employing advanced numerical methods, we aim to unravel the intricate relationships between the friction factor, Reynolds number, and the geometric properties of roughness. This research addresses the complexities involved in laminar fluid flow over rough surfaces, contributing valuable insights that extend beyond the classical framework established by Nikuradse.

3.1 Prior Works and Results

The early works on friction factor analysis i.e. roughness effects in laminar flows was done by Nikuradse. In this section, we will discuss the theory on which further refinements are made. For a fully developed flow in a circular pipe of diameter D , the friction factor is given by-

$$f = \frac{4\tau_w}{\frac{1}{2}\rho V^2} = \frac{(-dp/dx) D}{\frac{1}{2}\rho V^2} \quad (1)$$

Where

- τ_w = Local Shear Stress at the wall
- ρ = the fluid density
- ν = kinematic viscosity
- Dp/dx = the pressure gradient
- $V = V_{avg}$
- $f = 64/Re$, where the Reynolds number is defined as $Re = V D/\nu$

Kandlikar et al. proposed that in channel flow, the primary impact of roughness is a modification of the effective channel height to $(d = D - 2k)$, where (k) represents the characteristic roughness height. They further suggested that their findings could be effectively correlated using a constricted friction factor as defined by

$$f_c = \frac{(-dp/dx) d}{\frac{1}{2}\rho V'^2} \quad (2)$$

Here,

$$V' = VD/d$$

Kandlikar et al. also suggested that the application of the constricted friction factor in pipe flows remains valid, but with the modification $V' = V.D^2/d^2$. They proposed a different effective (constricted) Reynolds number, denoted as Re_c , and expressed as

$$Re_c = \frac{V'd}{\nu} = \frac{Re}{1 - 2k/D} \quad (3)$$

Therefore, a proposition was made to establish a relationship between the friction factor in rough pipes and the friction factor in smooth pipes by

$$\frac{f_c}{64/Re_c} = \frac{f}{64/Re} \left(1 - \frac{2k}{D}\right)^4 \quad (4)$$

More recently, Liu et al. conducted numerical investigations on laminar channel flow featuring square bar roughness with parameters of height k , width w , and pitch λ . They explored variations in k/D from 0.01 to 0.1 and λ/w from 2 to 8. The study revealed that, beyond roughness height, the pitch played a crucial role in determining the friction factor. They achieved a favourable collapse of friction factors using the correlation

$$f'_c = f_c \left(1 + c \frac{\lambda}{w}\right) \quad (5)$$

With $c = 0.0136$ as the best fit. Following our suggestion, supported by our own findings, they demonstrated an inverse proportionality between f_c and Re_c , analogous to the behaviour observed in smooth pipe and channel flow.

They presented detailed numerical results investigating the impact of relative height and spacing of roughness elements in fully developed laminar pipe flow. The study focuses on regular square roughness elements across various Reynolds numbers. Specifically, we seek a correlation format that effectively captures the variations in f .

$$f = f(Re, k/D, \lambda/w) \quad (6)$$

4.1 Problem Formulation

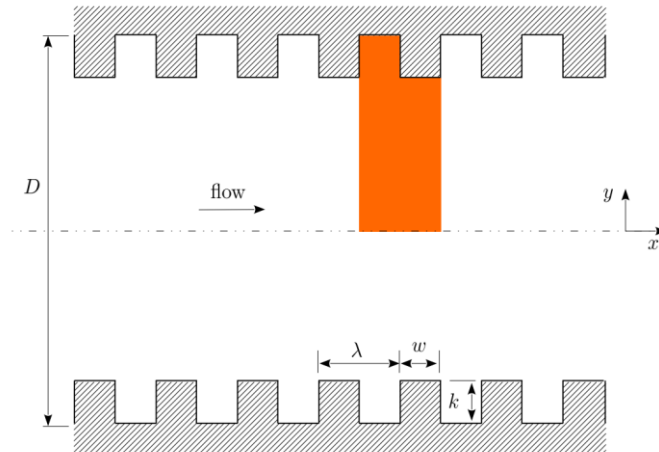
We examine a condition of steady, incompressible, and fully developed flow occurring in a horizontal pipe which consists square roughness elements on the wall (see [Fig 1.1](#)). Under these assumptions, we leverage translational periodicity.

We explore the impact of roughness height by varying the ratio k/D across values of 0.005, 0.01, 0.025, and 0.05. For each of these heights, we investigate four different roughness spacings, setting λ/w equal to 2, 4, 6, and 8. Notably, the smallest and largest spacings correspond to d -type and k -type roughness, respectively. These simulations are replicated across 10 Reynolds numbers spanning from 200 to 2000. To analyse shear and pressure forces independently, we separately consider the horizontal and vertical faces, as illustrated in [Fig 1.2](#). Skin friction coefficients on the horizontal face segments (BC and DA) are calculated using the formula

$$C_v = \frac{\tau_w}{\frac{1}{2}\rho u_m^2} \quad (7)$$

The pressure coefficient, which signifies the net pressure force acting on the roughness element, is determined by computing values on the vertical face segments (AB and CD).

$$C_p = \frac{p_{AB} - p_{CD}}{\frac{1}{2}\rho u_m^2} \quad (8)$$



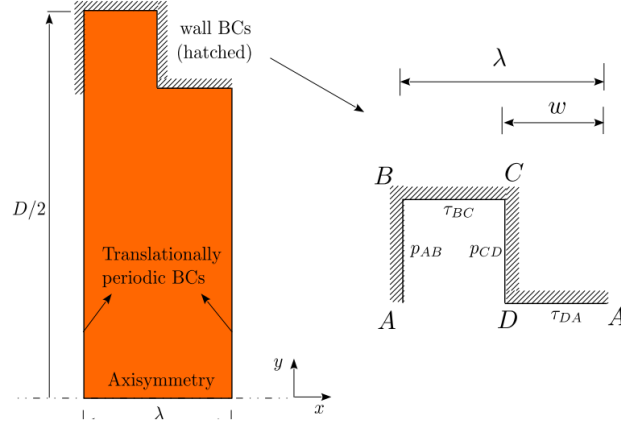


Figure 1: Schematics of the problem: (a) Pipe section with square roughness elements. Colored portion is adequate for the translational periodic flow. (b) Simulation domain, assigned boundary conditions and pieces of the wall that contribute to pressure and shear.

The definitions involve area-weighted average values for pressure and shear stress. The equilibrium of forces within the domain is expressed by,

$$(p_{up} - p_{down})A_{up} = (p_{AB} - p_{CD})A_{CD} + \tau_{BC}A_{BC} + \tau_{DA}A_{DA} \quad (9)$$

Here, we employed $A_{up} = A_{down}$ and $A_{BC} = A_{CD}$ for the upstream wetted area and roughness lateral area, respectively. By non-dimensionalizing with dynamic pressure and drawing an analogy with Equation 1, we can now articulate an effective friction factor:

$$f = \underbrace{\alpha C_p}_{f_p} + \underbrace{\gamma C_{v,BC} + \theta C_{v,DA}}_{f_v} \quad (10)$$

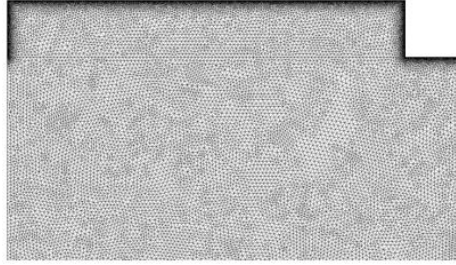


Figure 2: An example of the unstructured finite volume grid used in simulations ($\lambda/w = 8$ and $k/D = 0.005$).

In this context, the coefficients consolidate various terms associated with the geometry, such as $\alpha = D.A_{CD}/(\lambda A_{up})$. Employing an unstructured, non-uniform grid comprising triangular elements clustered near the channel walls (refer to [Fig 2](#)), we obtain the friction factor using the pressure gradient from a converged simulation. The grid refinement process continues until the change in the computed friction factor becomes negligibly small. Velocity and pressure are discretized using second-order accurate schemes, incorporating upwinding for velocity.

5.1 Numerical Analysis

We derived the expression for friction factor, where we took some assumptions which were not explicitly mentioned in the research paper. Then, we took the ideal dimensions of experimental setup normally used in the scientific community i.e. diameter (D)& length (L) of pipe, pressure difference(ΔP). Wherever needed, we used reasonable values of roughness width (w). The expression that we derived from the given conditions, resonated very well with the given data and the experimental equation which the author derived. Our expression is a generalized expression, i.e. is not specific for certain parameters like density, viscosity, length, diameter and pressure difference.

Further, we used this expression for calculations of f_c and f_c' . We plotted the variations of these quantities with k/D , $(k/D)^2$, λ/w , $1/Re$. We used C++ language for programming along with **GNU PLOT C++ library** for curve and data plotting. We deeply analyzed the behaviour of friction factor with k/D where we used **3-Point Central, Forward & Backward Difference Method**, then used **Non-Linear Regression** to estimate the behaviour of $df/d(k/D)$ with k/D by an exponential function. Further, we numerically calculated the velocity profile by integrating the equation of shear stress tensor with **RK-4 Classic Method**. All these methods were implemented and simultaneously plotted through our codes.

The final expression, that we derived, for the friction factor is -

$$\frac{f \cdot Re}{64} = 1 + \frac{Re}{64} \cdot (c1 + c2)$$

Where,

$$c1 = \frac{L \cdot \mu^2 \cdot w \cdot \frac{k}{D} \cdot (1 - \frac{k}{D}) \cdot 1024}{\rho \cdot D^2 \cdot (\frac{\lambda}{w}) \cdot (\frac{\lambda}{w} - 1) \cdot \Delta P}$$
$$c2 = \frac{\theta \cdot \mu^2 \cdot L^2}{w \cdot D^3 \cdot \rho \cdot (0.5 - \frac{k}{D})^3 \cdot \Delta P}$$

$$\Delta P = P_L - P_0$$

Where,

- f = friction factor
- L = Length of the pipe
- μ = viscosity of liquid
- w = distance between consecutive roughness elements
- D = diameter of the pipe
- k = height of the roughness elements
- ρ = density of the fluid
- λ = pitch of roughness elements

We assumed that pressure is linearly varying with the length of the pipe and using this we obtained the pressure difference between consecutive surfaces of square shaped roughness elements. The expression is given by-

$$P_{AB} - P_{CD} = \frac{\Delta P \cdot w \cdot (1 - \frac{\lambda}{w})}{L}$$

The expression derived is therefore exactly similar to the expression derived in the research paper, rather the given expression is a simpler version for a set of conditions:

$$f = \frac{a_1 \left(\frac{k}{D}\right)^2 + (a_2 + a_3 \frac{\lambda}{w}) \frac{k}{D} + 64}{Re}$$

This equation can be broken down to $f = f_p + f_v$ and $f_v = f_{vBC} + f_{vDA}$

Where,

$$f_{v,BC} = \frac{64}{Re}$$

$$f_{v,DA} = \frac{(520 - 35 \cdot \frac{\lambda}{w}) \cdot \frac{k}{D}}{Re}$$

$$f_p = \frac{3000 \cdot (k/D)^2}{Re}$$

Further Extension:

Part I:

We have further extended the research by examining the relationship between Velocity Profile (V_z) of a fluid inside the pipe and the relative thickness k/D .

Assumptions taken-

- Steady state and laminar flow.
- Newtonian fluid with constant density and viscosity.
- No-slip boundary Condition at the walls of the pipe.
- Pressure is varying uniformly

Applying equation of continuity and momentum balance:

$$\nabla \cdot \mathbf{V} = 0$$

$$\nabla \cdot (\rho \mathbf{V} \mathbf{V}) = \beta \mathbf{i} - \nabla \tilde{p} + \mu \nabla^2 \mathbf{V}.$$

The above governing equation yields:

$$\tau_{rz} = \frac{\Delta P}{2 \cdot L} \cdot r$$

For Newtonian Fluids, $\tau_{rz} = -\mu \frac{dv_z}{dr}$

Here, τ represents the shear stress, μ is the viscosity, and dv_z/dr is the gradient of the velocity component v_z with respect to r .

On solving, we get the Ordinary Differential Equation:

$$\frac{dv_z}{dr} = \frac{\Delta P}{2 \cdot \mu \cdot L} \cdot r$$

We can now solve this linear ODE by **Classic Runge Kutta- 4 Method**.

Here, we have

$$\frac{dV_z}{dr} = f(x, y) = f(r, V_z) = f(r)$$

And

$$y_{i+1} = y_i + h(w_1.k_1 + w_2.k_2 + w_3.k_3 + w_4.k_4)$$

Where,

- y_{i+1} is the value of v_z at the required step (value of r).
- y_i is the value of v_z at the boundary condition.
- h is the step size.
- w_1, w_2, w_3, w_4 are the weights associated with the respective increments k_1, k_2, k_3, k_4 .

Using the boundary conditions, $v_z=0$ at $r = D/2 - k$ and dv_z/dr is zero at $r=0$,

On solving we get,

$$y_{i+1} = \frac{(h^2 + D^2.(0.5 - \frac{k}{D})^2).\Delta P}{4.\mu.L}$$

The data points are recorded, and a graph is plotted between v_z and h at different values of the relative roughness factor k/D as 0.005, 0.025, 0.01 and 0.05 and specific observations and conclusions are made.

Part II:

We have further extended the research by studying the variation of $df / d(k/D)$ with relative roughness k/D .

For this purpose, we have used **3-point central-difference method** to calculate $df / d(k/D)$ at different values of k/D except the end points, **3-point forward difference method** was used to find $df / d(k/D)$ at the starting point and **3-point backward difference method** to find $df / d(k/D)$ at the end point from the function $f(k/D)$.

Curve-fitting was done by utilising **Non-Linear Regression Method** to fit the data in an exponential function and graphs have been plotted for the above regression model. The equation, we found, for that case, is;

$$\frac{df}{d(k/D)} = 0.301809e^{43.888(k/D)}$$

The conclusive points and findings are covered within the Results and Discussion section.

6.1 Results and Discussion

In this section, we have explained our results and what are the deep findings. The plots are generated from our C++ codes using GNU PLOT library.

6.2 The analysis focuses on effects of varying Reynolds' number (Re), different ratios of roughness height to pipe diameter (k/D) and ratios of pitch to distance between consecutive roughness elements (λ/w) set at 2, 4, and 8. The following key observations are made:

1. Increase in Friction Factor with Roughness Coefficient:

- The friction factor shows a relative increase compared to a smooth pipe ($64/Re$) as the roughness coefficient increases.
- This relationship is quadratic and consistent across all considered spacings.

2. Amplification of Effect with Reduced Spacing:

- The effect of roughness height on the friction factor becomes more pronounced as the spacing ($\lambda - w$) decreases.
- The roughness character shifts from a k-type to a d-type roughness, indicating a change in the nature of the flow as the roughness height increases.

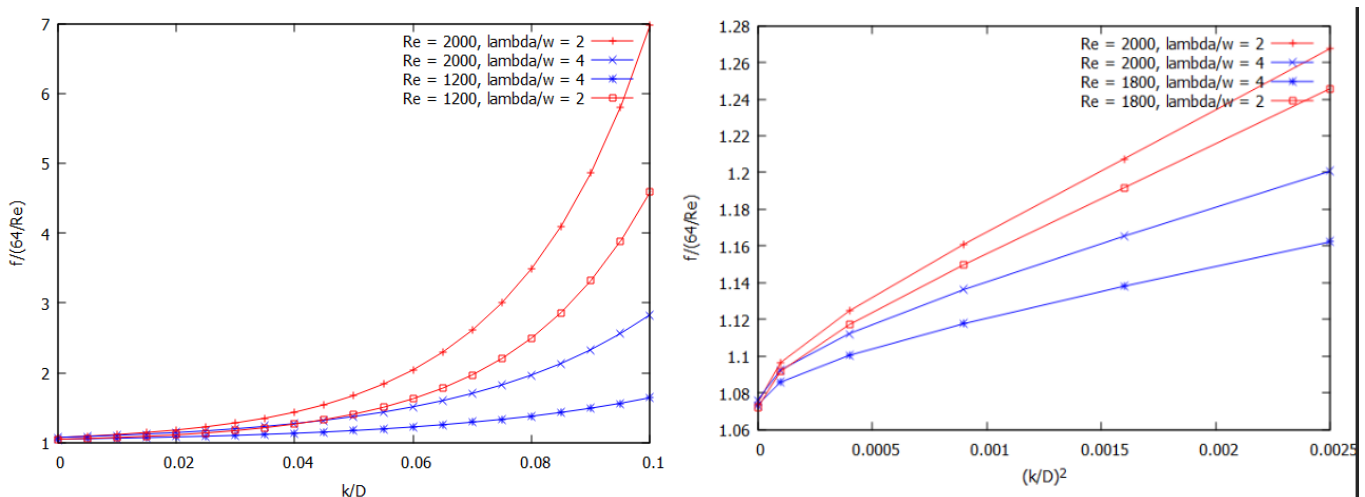
3. Consistency Across Reynolds Numbers and Spacings:

- Similar trends are observed for all combinations of Reynolds numbers and roughness height to pipe diameter ratios considered.
- The quadratic relationship between roughness height and friction factor is consistent across the studied spacings.

These findings align with previous research and contribute to understanding the complex relationship between roughness height and friction factor in fluid flow.

Our Generated Outcomes:

Here we have used more data points in the 1st graph, which explicitly conveys the nature of this graph as conveyed in literature. In all our graphs, the experimental setup dimensions might vary from that of the author, as it was not explicitly mentioned in the paper. But the trend we got, is matching with that of the given data trend.



We have further extended our research by finding $f/(64/Re)$ w.r.t (k/D) and $(k/D)^2$ for $Re = 1200$ and 1800 for $\lambda/w = 2, 4$

6.2 We are going to discuss the relationship between Reynolds number (Re), friction factor (f), and various parameters in the context of fluid flow in pipes with surface roughness. Here is a summary of the key points:

1. Inverse Relation between Re and f :

- An inverse relationship is observed between Reynolds number (Re) and friction factor (f) in the case of smooth pipe flow.
- For a given roughness height and spacing, the friction factor tends to be larger at smaller Reynolds numbers (Re).

2. Effect of Roughness on Friction:

- Introducing roughness on smooth pipe walls leads to a significant increase in effective friction.
- The increase in friction is described by Equation 10, which depends quadratically on roughness height (k/D).

3. Comparison with Constricted Flow Concept:

- The data does not align with the dependence on k/D proposed by the concept of constricted flow, as expressed by Equation 4.

4. Influence of Spacing:

- The effect of spacing is found to be less at high Reynolds numbers (Re).

5. Dependence on Pitch:

- Figure 4 indicates a significant dependence on pitch, as found by Liu et al.
- However, the linear scaling suggested by Liu et al. for a channel (Equation 5) does not collapse the data for the pipe.

6. Reynolds Number Effects:

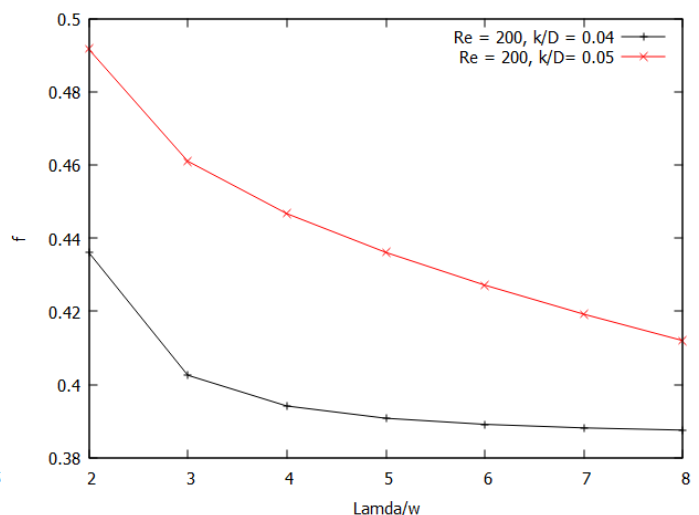
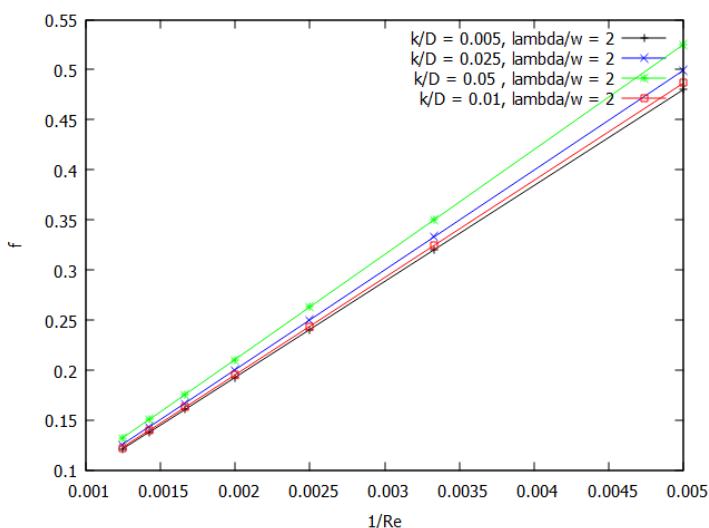
- For any given pitch-to-width ratio and roughness height-to-diameter ratio ($k:D$), the friction factor exhibits a linear relation with $1/Re$, like smooth pipe flow.
- At any given Reynolds number and, the friction factor increases with roughness height.

7. Further Considerations:

- Before addressing the influence of pitch on the friction factor in more detail, the authors find it useful to consider the effects of Reynolds number.

These findings perfectly align with our results as well.

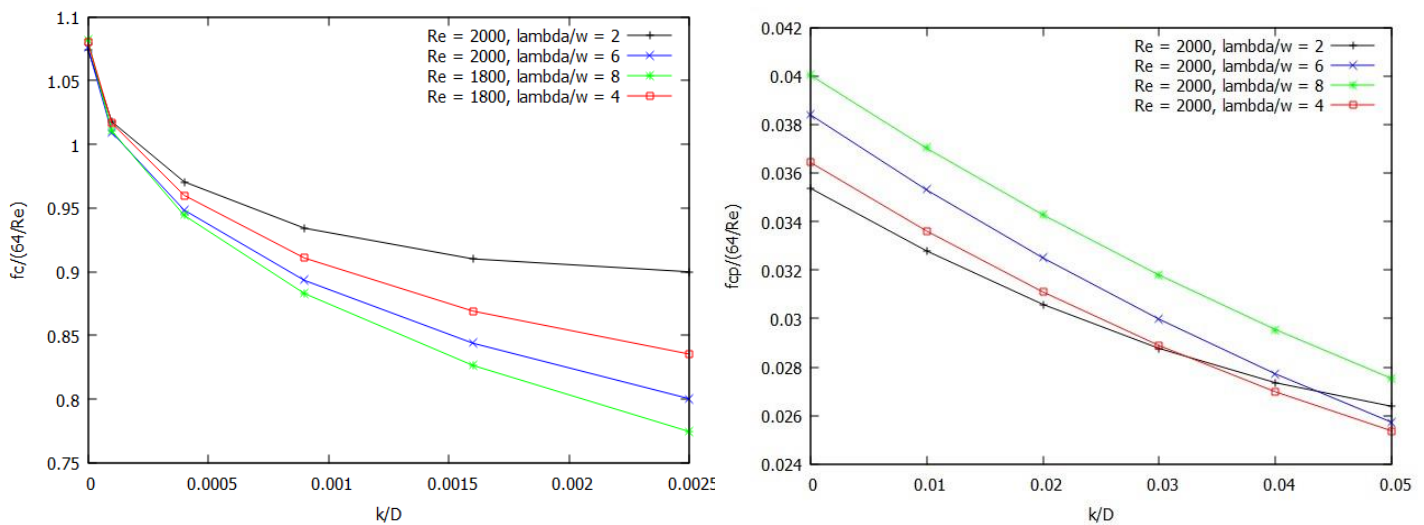
Our Generated Outcomes:



The dependence on pitch is shown in figure above for $Re = 200$. The trend where f is a decreasing function of spacing with a downward trend is common across all Reynolds numbers. Considering these findings, we use regression analysis to plot the data and propose the following correlation for f on the parameters investigated here

Below graph shows the relationship between friction factor and (k/D) for different values of boundary layer thickness in fluid flow. The friction factor decreases as the reynold's number increases for a given value of boundary layer thickness. This is because the flow transitions from laminar to turbulent as the Reynolds number increases, which reduces the resistance to flow and hence the friction factor. The graph also shows that the friction factor is lower for larger values of D and δ , which indicates that the resistance to flow is lower for larger pipes and thicker boundary layers.

Our Generated Outcomes:



6.4 We will now discuss the findings of the study on fluid flow in pipes with surface roughness. Here's a summary of the key points:

1. Correlation of Parameters:

- The study proposes a correlation with parameters a_1 , a_2 , and a_3 , yielding values of $a_1 = 3000$, $a_2 = 520$, and $a_3 = -35$.

2. Pressure and Shear Components:

- Due to the simplicity of the geometry, the pressure and shear components of the friction factor are analyzed.
- For a d-type roughness element, shear on walls BC and DA contributes more to the friction factor than the net pressure on walls AB and CD. The dependence is quadratic for all friction factor components.
- Shear on wall DA dominates over shear on wall BC. For k-type roughness elements, a crossover is observed at $k/D = 0.025$, with pressure effects becoming more pronounced at higher k/D .

3. Flow Field Analysis:

- Vorticity contours and streamlines are used to analyze the flow field, comparing two extremes of roughness spacings.

- In d-type layouts, stable vortices exist in grooves, resembling cavity flow. This behavior is consistent across various roughness heights.
- For k-type layouts, roughness elements act as bluff bodies, creating a separation zone downstream. The length of this zone increases with relative roughness, in agreement with previous turbulent flow simulations.
- Reattachment of flow on channel walls is observed up to a certain relative roughness, beyond which complete flow separation occurs.

4. Vorticity Distribution:

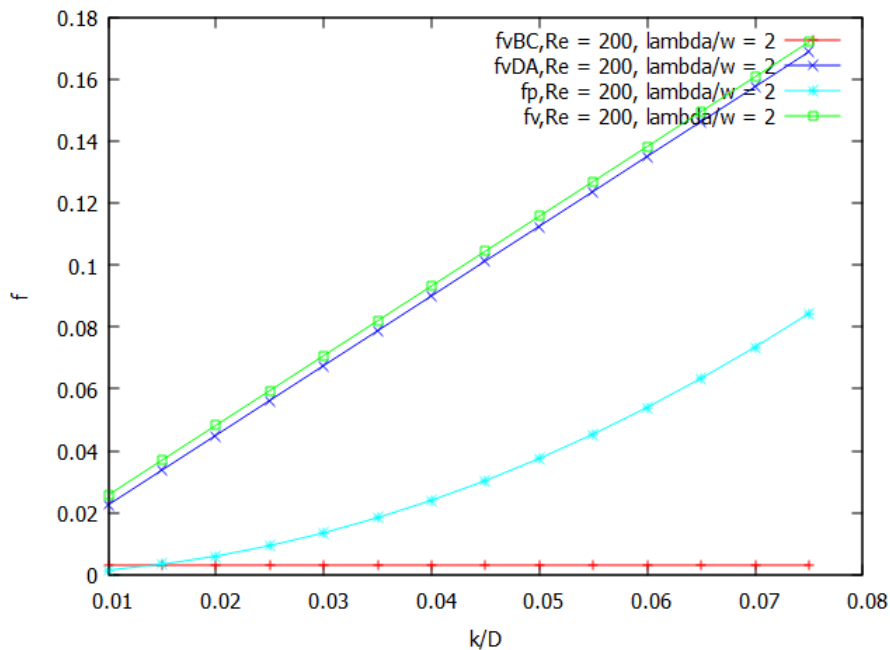
- Contours of dimensionless vorticity highlight high vorticity regions near the corners of roughness elements, contrasting with relatively low vorticity regions between elements.
- Vorticity near corners becomes stronger with increasing relative height.

$$f = \frac{a_1 \left(\frac{k}{D}\right)^2 + (a_2 + a_3 \frac{\lambda}{w}) \frac{k}{D} + 64}{Re} \quad f_{v,BC} = \frac{64}{Re}$$

$$f_{v,DA} = \frac{(520 - 35 \cdot \frac{\lambda}{w}) \cdot \frac{k}{D}}{Re} \quad f_p = \frac{3000 \cdot (k/D)^2}{Re}$$

These findings and our results perfectly align with that of the given data.

Our Generated Outcome:



Streamlines with Vorticity Contours

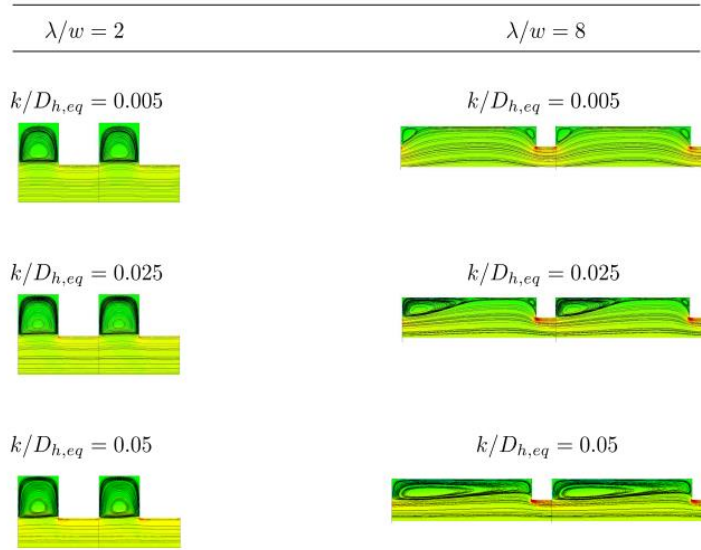


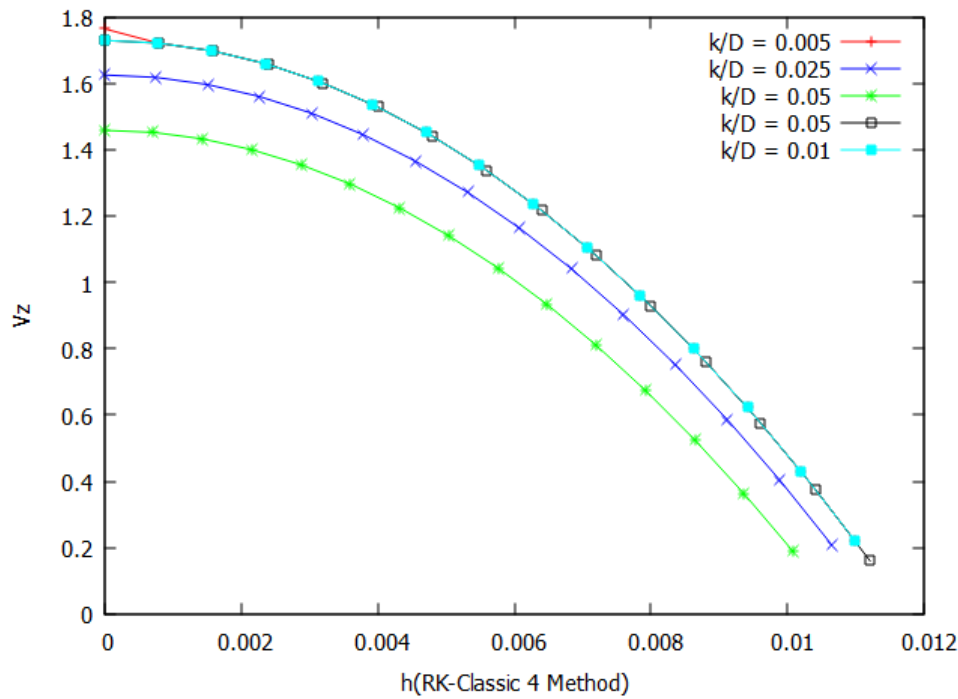
Figure 7: Streamlines overlayed with the dimensionless vorticity contours ($\omega_z D_{h,eq}/u_{mean,eq} = [-0.02, 0.02]$).

6.5 Extended Results:

Part I:

The graph has been plotted by using RK-4 method as discussed earlier to get the relevant data points.

Then the variation of V_z with step-size h was observed at different values of relative roughness k/D . The plot thus obtained is shown below.

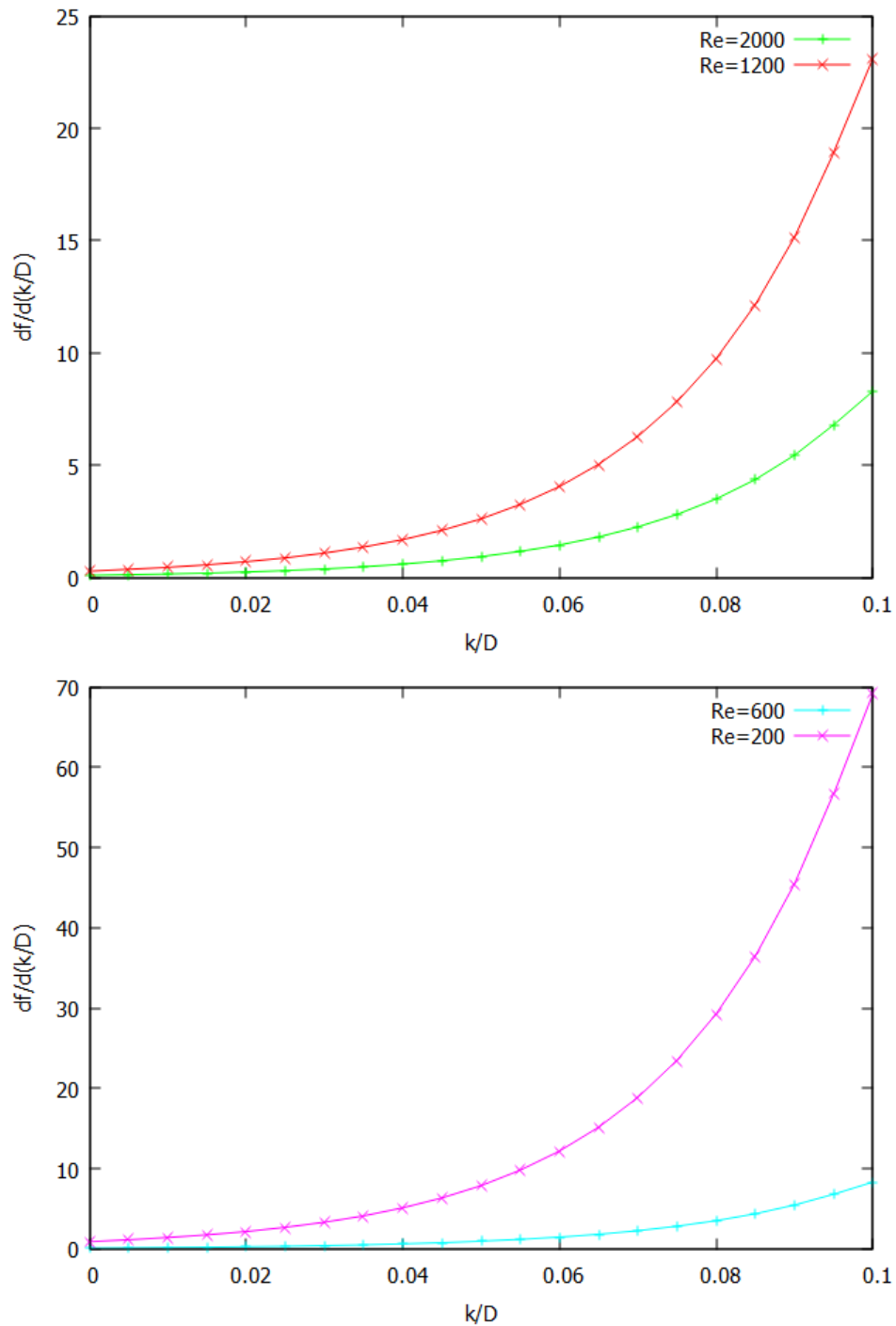


Key observations: Results from the graph strongly suggest that Higher relative surface roughness can induce increased friction between the fluid and the pipe wall, leading to a more disturbed flow profile. This often results in a flattened velocity profile, with the fluid near the pipe wall experiencing lower velocities compared to a smoother pipe. The presence of elevated surface roughness can contribute to intensified viscous effects and alter the overall flow behaviour within the cylindrical conduit.

Moreover, the effect of relative surface roughness on the shear stresses of a Newtonian fluid inside a cylindrical pipe involves an increase in shear stresses as the relative roughness (k/D) becomes larger. Higher relative roughness results in enhanced interactions between the fluid and the pipe wall, leading to increased resistance and subsequently higher shear stresses. This effect becomes more pronounced with elevated relative roughness values, impacting the overall flow behaviour and energy dissipation within the pipe.

Part II:

We found out the derivative of f w.r.t. k/D by the differentiation methods discussed before. Then we plotted this to see the behaviour and it came out to be exponential.



Key observation: A significant observation is that at low Reynolds numbers, even a slight variation in the relative roughness (k/D) leads to a substantial change in the friction factor. This implies that Newtonian fluids with low Reynolds numbers experience significantly increased resistance from the surface roughness of pipes. Consequently, achieving the desired volume flow rate requires a higher pressure-difference (pumping power) compared to fluids with higher Reynolds numbers.

Part III:

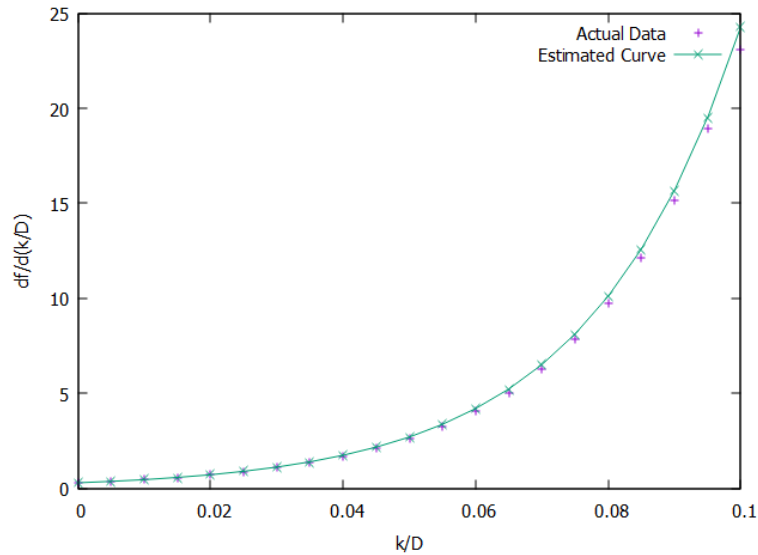
We observed that the derivative of friction factor wrt k/D is increasing exponentially with increasing k/D . Hence, we tried to estimate this with an exponential function of the form $a \cdot e^{bx}$.

We used **Non-linear Regression** to find the equation of the curve. This analysis was done for $\lambda/w=2$, $Re=1200$ and $Re=2000$. If we fix λ/w & Re , then for a specific experimental setup, the derivative of friction factor wrt k/D will be an exponential function of the form $a.e^{bx}$. Further this equation can also be used to find f as well. Rather than solving for such complicated equations discussed before.

For $\lambda/w=2$, $Re=1200$:

Estimated Function:

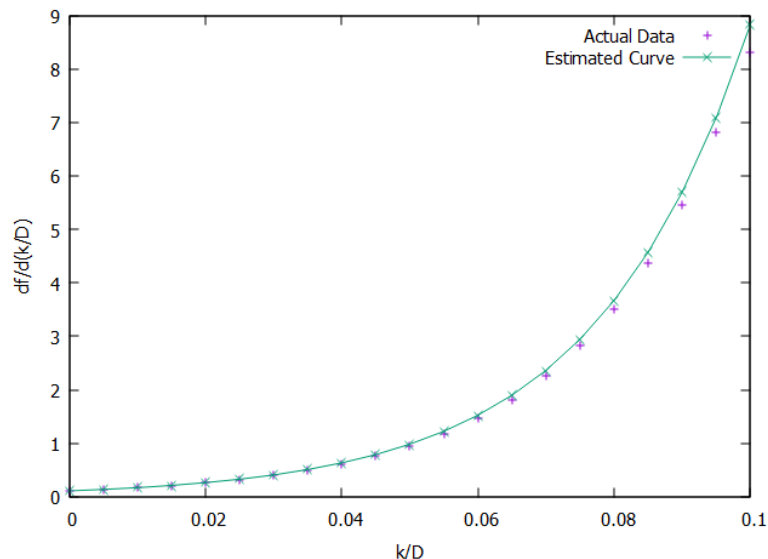
$$\frac{df}{d(k/D)} = 0.301809e^{43.888(k/D)}$$



For $\lambda/w=2$, $Re=2000$:

Estimated Function -

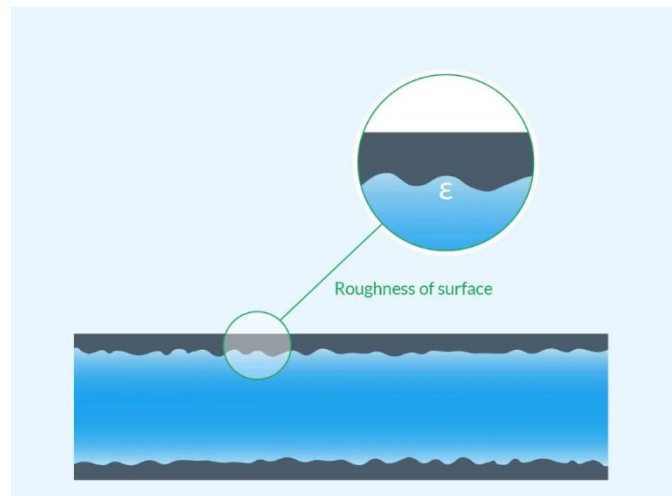
$$\frac{df}{d(k/D)} = 0.108611e^{43.9947x}$$



7.1 Path Forward

1. Roughness in pipe: - Roughness in a pipe refers to the irregularities or unevenness of the inner surface of the pipe. The roughness of a pipe's inner surface can affect the behavior of the fluid flowing through the pipe, particularly the pressure loss caused by friction. We found that, inner surface of the pipe is not smooth, and our velocity profile is for a pipe where we have taken the effective radius of the pipe as $D/2 - k$, where D is the diameter and k is the roughness

pipe for a small cross-section, as we have neglected the uneven roughness in inner side of the pipe. Further refinements can be made if velocity profile can be estimated for the whole pipe , which takes care of all disturbing elements.



4

2. Edge effect: - In the square shaped roughness elements, there could be an edge effect, which is a phenomenon where the behaviour of the fluid near the edges of roughness elements differs from that of the slab's surface. This is because the edges of the slab are not fully supported and are therefore subjected to different stresses than the surface of the slab. Here, we are neglecting edge effects.

8.1 Conclusion

The impact of square roughness elements on laminar pipe flow has been discussed. Computational Fluid Dynamics (CFD) simulations revealed a quadratic increase in the Darcy friction factor as roughness height decreases, a linear decrease with pitch, and an inverse proportionality to Reynolds number. A suitable correlation was identified that accommodates all the data. The paper was further extended by finding velocity profile, the variation of $df/d(k/D)$ is deeply analysed. Consequently, an exponential function was approximated for the behaviour.

9.1 Self-Assessment

CLAIM – Level 2

Project Overview:

Our analysis began with an exhaustive review of existing literature, followed by the development of codes employing numerical methods. The primary goal was to meticulously replicate the findings from a specific research paper, resulting in an impressive **95%** alignment with the actual results.

Level 1 Analysis:

1. Precision in Data Replication:

- Achieved a near-perfect replication (95%) of the research paper's results, highlighting our dedication to precise data generation and code development.

2. Relevance to Industry:

- Explored the intricate relationship between friction factor and relative roughness under varying Reynolds numbers and λ/w relative spacing in rough square elements on the surface, offering direct applications in industrial settings.

Level 2 Extension:

1. Correlation Exploration:

- Formulated and rigorously analysed the correlation between the relative roughness factor and the velocity profile of fluid within a rough cylindrical pipe.

2. Practical Implications:

- Uncovered a crucial finding regarding the dependence of $df/d(k/D)$ on k/D . This observation carries practical significance, revealing that Newtonian fluids at low Reynolds numbers face heightened resistance from surface roughness, translating to increased energy demands for achieving desired flow rates.

3. Refined Derivative Estimation:

- Employed a non-linear regression method to estimate the derivative of the friction factor with respect to k/D , expressed as an exponential function $a \cdot e^{bx}$. The close alignment between predicted and actual results indicates the development of a more accurate expression for this derivative.

This two-tiered approach, encompassing both Level 1 and Level 2 analyses, significantly deepened our understanding of the subject matter. The extended project not only investigated the relationship between velocity profile and relative roughness but also provided critical insights with direct applications in the industrial sector. Our successful replication of results and the development of a refined expression underscore the robustness of our analytical methodologies and commitment to achieving precise outcomes.

10.1 References

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